

Education

2027 (expected) Department of Medicine, School of Medicine, International University of Health and Welfare, Chiba, Japan (M.D., expected March 2027)

Appointments and Experience

- 2026–present** Academic Visitor, Saw Swee Hock School of Public Health, National University of Singapore, Singapore (Supervisor: Assistant Professor Akira Endo)
- Network modelling for sexual transmitted infections
- 2026** Clinical Electives Student, Division of Dermatology, National University Hospital, Singapore (Supervisor: Associate Professor Nisha Suyien Chandran)
- Completed a four-week clinical elective in dermatology.
 - Gained clinical exposure to outpatient consultations, inpatient care and dermatological procedures.
- 2024–present** Lab Associate, SPARK Lab NYC, Environmental Health Sciences, Columbia University, New York, USA (Supervisor: Assistant Professor Robbie M. Parks)
- Spatio-temporal hierarchical modelling of the mortality impacts of tropical cyclones and climate change.
- 2020–2024** Advisor, National COVID-19 Cluster Task Force, Ministry of Health, Labour and Welfare, Tokyo, Japan (Supervisor: Professor Hiroshi Nishiura)
- Conducted ad-hoc analyses to support risk assessment of the COVID-19 epidemic and to evaluate the impact of public health responses.
 - Conducted analyses on Alpha-variant dynamics, vaccine effectiveness against death using population-level data, and the impact of healthcare burden on temporal case fatality risk (e.g. Murayama et al. Theoretical Biology and Medical Modelling. 2021; Ko et al. NIID. 2021; Ko et al. Emerging Infectious Diseases. 2022).
- 2022–2023** Research Assistant, Graduate School of Public Policy, The University of Tokyo, Tokyo, Japan (Supervisor: Associate Professor Taisuke Nakata)
- Collaborated with economists on research into COVID-19 response strategies.
- 2021–2023** Member, CoV-Navi (<https://covnavi.jp/>)
- Reviewed scientific evidence on COVID-19 vaccines for public-facing science communication.
- 2021–2022** Research Assistant, Graduate School of Social Sciences, Chiba University, Chiba, Japan (Supervisor: Assistant Professor Shouto Yonekura)
- Developed a novel Bayesian framework for estimating waning vaccine effectiveness from population-level surveillance data in the presence of multiple circulating variants, in collaboration with Dr Akira Endo (Murayama et al. Epidemics. 2023).

Selected Publications

1. [Murayama H](#)[†], Asakura TR[†], D Dickens BL, Boyle D, Foo JH, Jin S, Mukadi PK, Ejima K, Jung S-m, Nishi A, Prem K, Wakamba AM, Saila-Ngita D, Niyukuri D, Endo A. Role of community and sexual contacts as drivers of MPXV clade I. *Nature Health*. 2026 Apr 6. doi:10.1038/s44360-026-00080-7
2. [Murayama H](#), Endo A, Yonekura S. Estimation of waning vaccine effectiveness from population-level surveillance data in multi-variant epidemics. *Epidemics*. 2023;100726.
3. [Murayama H](#), Pearson CAB, Abbott S, Miura F, Jung S, Fearon E, Funk S, Endo A. Accumulation of immunity in heavy-tailed sexual contact networks shapes mpox outbreak sizes. *The Journal of Infectious Diseases*. 2023 Jul 4;jjad254.
4. Endo A, [Murayama H](#), Abbott S, Ratnayake R, Pearson CAB, Edmunds WJ, Fearon E[†], Funk S[†]. Heavy-tailed sexual contact networks and monkeypox epidemiology in the global outbreak, 2022. *Science*. 2022 Sep 25;0(0):eadd4507.

5. Ko Y, [Murayama H](#), Yamasaki L, Kinoshita R, Suzuki M, Nishiura H. Age-dependent effects of COVID-19 vaccination and healthcare burden on COVID-19 deaths in Tokyo, Japan. *Emerging Infectious Diseases*. 2022;28(9).

Publication List

Peer-Reviewed Original Research (†: equal contribution)

1. [Murayama H](#)†, Asakura TR†, Dickens BL, Boyle D, Foo JH, Jin S, Mukadi PK, Ejima K, Jung S-m, Nishi A, Prem K, Wakamba AM, Saila-Ngita D, Niyukuri D, Endo A. Role of community and sexual contacts as drivers of MPXV clade I. *Nature Health*. 2026 Apr 6. doi:10.1038/s44360-026-00080-7
2. Jin S, Asakura TR, [Murayama H](#), Jung S, Niyukuri D, Nyandwi J, Nkengurutse L, Kamatari O, Lim JT, Endo A, Dickens BSL. Disentangling temporal trends of clade Ib monkeypox virus transmission in Burundi. *The Journal of Infectious Diseases*. 2025 Sep 10;jiaf475.
3. Jin S, Asakura TR, [Murayama H](#), Niyukuri D, Saila-Ngita D, Lim JT, Endo A, Dickens BSL. Vaccination strategies to achieve outbreak control for MPXV clade I with a one-time mass campaign in sub-Saharan Africa: a scenario-based modelling study. *PLOS Medicine*. 2025 Sep 5;22(9):e1004726.
4. Ejima K†, Wang Y†, Endo A†, [Murayama H](#), Goh YS, Cook AR, Jeong YD, Iwami S, Park H, Dickens BSL, Jin S, Lim JT, Chan CEZ, Chia PY, Young BE, Chio M, Lye DC, Ajelli M. Evaluating the effectiveness of international travel controls to identify MPXV-infected travellers: a simulation study. *BMC Medicine*. 2025 Aug 12;23(1):473.
5. Asakura TR, Jung S, [Murayama H](#), Ghaznavi C, Sakamoto H, Teshima A, Miura F, Endo A. Modelling international spread of clade IIb mpox on the Asia continent. *Bulletin of the World Health Organization*. 2025;103(7):429-436.
6. Jung S†, Miura F†, [Murayama H](#), Funk S, Wallinga J, Lessler J, Endo A. Dynamic landscape of mpox importation risks driven by heavy-tailed sexual contact networks among men who have sex with men in 2022: a mathematical modelling study. *The Journal of Infectious Diseases*. 2024;jiae433.
7. [Murayama H](#), Endo A, Yonekura S. Estimation of waning vaccine effectiveness from population-level surveillance data in multi-variant epidemics. *Epidemics*. 2023;100726.
8. [Murayama H](#), Pearson CAB, Abbott S, Miura F, Jung S, Fearon E, Funk S, Endo A. Accumulation of immunity in heavy-tailed sexual contact networks shapes mpox outbreak sizes. *The Journal of Infectious Diseases*. 2023 Jul 4;jiad254.
9. Endo A, [Murayama H](#), Abbott S, Ratnayake R, Pearson CAB, Edmunds WJ, Fearon E†, Funk S†. Heavy-tailed sexual contact networks and monkeypox epidemiology in the global outbreak, 2022. *Science*. 2022 Sep 25;0(0):eadd4507.
10. Ko Y, [Murayama H](#), Yamasaki L, Kinoshita R, Suzuki M, Nishiura H. Age-dependent effects of COVID-19 vaccination and healthcare burden on COVID-19 deaths in Tokyo, Japan. *Emerging Infectious Diseases*. 2022;28(9).
11. [Murayama H](#)†, Yamasaki L†, Hashizume M. The impact of temperature on the transmissibility and virulence of COVID-19 in Tokyo, Japan. *Scientific Reports*. 2021;11(1):24477.
12. [Murayama H](#), Kayano T, Nishiura H. Estimating COVID-19 cases infected with the Alpha variant (VOC 202012/01): an analysis of screening data in Tokyo, January–March 2021. *Theoretical Biology and Medical Modelling*. 2021;18(1):13.

Journal Correspondence

1. Jung S, Miura F, [Murayama H](#), Funk S, Wallinga J, Lessler J, Endo A. Preemptive Mpox Vaccine Deployment: Aligning Strategy with Reality. *The Journal of Infectious Diseases*. 2025 Jul 21;jiaf365.

Governmental reports

1. Ko KY, [Murayama H](#), Yamasaki L, Kinoshita R, Nishiura H, Suzuki M. Evaluating the age-specific effectiveness of COVID-19 vaccines against death from surveillance data in Tokyo. *National Institute of Infectious Diseases, Infectious Diseases Surveillance Center*. 2021 Dec. <https://id-info.jihs.go.jp/niid/ja/2019-ncov-e/10873-covid19-65.html> (in Japanese).
2. Ko KY, [Murayama H](#), Yamasaki L, Kinoshita R, Nishiura H, Suzuki M. Evaluating the age-specific effectiveness of COVID-19 vaccines against death from surveillance data in Tokyo. *Materials 3-2, 65th Advisory Board Meeting, Ministry of Health, Labour and Welfare on COVID-19 Countermeasures* (28 December 2021), 79-90. 28 Dec 2021. <https://www.mhlw.go.jp/content/10900000/000875165.pdf> (in Japanese).

Conference Presentation

1. [Murayama H](#), Endo A. Transmission dynamics and risk assessment of mpox clade IIB and IB within men who have sex with men. *Early Career Researcher Sandbox session, Infectious Disease Modelling conference*. 2024 Nov. Bangkok, Thailand. Oral presentation.
2. [Murayama H](#). Impacts of vaccine, healthcare burden, and temperature on the transmissibility or virulence of COVID-19. *COVID-19 Pandemic Conference*. 2022 Sep. Nagoya, Japan. Oral presentation.

Skills and Professional Development

Technical Expertise

- Research areas: (i) infectious disease epidemiology and mathematical modelling of infectious diseases (COVID-19, mpox clade I and II, sexually transmitted infections, dengue, vaccine effectiveness); (ii) environmental epidemiology, focusing on interactions between infectious disease dynamics, temperature, and tropical cyclones.
- Data analysis and scripting languages: R, Julia, Stan.
- Statistical computing environments: JupyterLab (Windows), Docker; familiar with RStudio.
- Other software: Microsoft Office, GitHub, Zotero.
- Markup languages: LaTeX, Markdown; familiar with HTML, XML, and CSS.
- Experience with Bayesian methods, maximum likelihood estimation, differential equations, stochastic processes, branching processes, spatio-temporal modelling, and network modelling.

Language

- Japanese (native)
- English (advanced)

Professional Services

Co-reviewer for peer-reviewed journals: PLoS Neglected Tropical Diseases (1), PLoS ONE (1), The Journal of Infectious Diseases (1).

Teaching Experience

July 2025	Teaching Assistant, Introduction to Infectious Disease Epidemiology and Modelling, School of Tropical Medicine and Global Health, Nagasaki University, Japan
July 2023	Teaching Assistant, Introduction to Infectious Disease Epidemiology and Modelling, School of Tropical Medicine and Global Health, Nagasaki University, Japan

Membership

1. Japan Epidemiological Association
2. Japanese Society of Tropical Medicine